

COVID_Analysis.ipynb

April 17, 2026

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

print(" All libraries imported successfully!")
```

All libraries imported successfully!

```
[3]: url = "https://raw.githubusercontent.com/owid/covid-19-data/master/public/data/
      ↪owid-covid-data.csv"
df = pd.read_csv(url)
df.head()
```

```
[3]: iso_code continent    location    date  total_cases  new_cases  \
0      AFG      Asia  Afghanistan  2020-01-05         0.0         0.0
1      AFG      Asia  Afghanistan  2020-01-06         0.0         0.0
2      AFG      Asia  Afghanistan  2020-01-07         0.0         0.0
3      AFG      Asia  Afghanistan  2020-01-08         0.0         0.0
4      AFG      Asia  Afghanistan  2020-01-09         0.0         0.0

      new_cases_smoothed  total_deaths  new_deaths  new_deaths_smoothed  ...  \
0                  NaN            0.0          0.0                  NaN  ...
1                  NaN            0.0          0.0                  NaN  ...
2                  NaN            0.0          0.0                  NaN  ...
3                  NaN            0.0          0.0                  NaN  ...
4                  NaN            0.0          0.0                  NaN  ...

      male_smokers  handwashing_facilities  hospital_beds_per_thousand  \
0             NaN                   37.75                        0.5
1             NaN                   37.75                        0.5
2             NaN                   37.75                        0.5
3             NaN                   37.75                        0.5
4             NaN                   37.75                        0.5
```

	life_expectancy	human_development_index	population	\
0	64.83	0.51	41128772	
1	64.83	0.51	41128772	
2	64.83	0.51	41128772	
3	64.83	0.51	41128772	
4	64.83	0.51	41128772	

	excess_mortality_cumulative_absolute	excess_mortality_cumulative	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

	excess_mortality	excess_mortality_cumulative_per_million
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

[5 rows x 67 columns]

```
[4]: print("Shape of data (rows, columns):", df.shape)
print("\nFirst 15 column names:\n", df.columns.tolist()[:15])
df['date'] = pd.to_datetime(df['date'])
key_cols = ['new_cases', 'total_cases', 'new_deaths', 'total_deaths',
            'new_vaccinations', 'total_vaccinations']
for col in key_cols:
    if col in df.columns:
        df[col] = df[col].fillna(0)

print("\nDate converted + missing values filled with 0 in key columns!")
print("Missing values left in main columns:\n", df[['date', 'location',
            'new_cases', 'total_cases', 'new_deaths', 'total_deaths']].isnull().sum())
```

Shape of data (rows, columns): (429435, 67)

First 15 column names:

```
['iso_code', 'continent', 'location', 'date', 'total_cases', 'new_cases',
'new_cases_smoothed', 'total_deaths', 'new_deaths', 'new_deaths_smoothed',
'total_cases_per_million', 'new_cases_per_million',
'new_cases_smoothed_per_million', 'total_deaths_per_million',
'new_deaths_per_million']
```

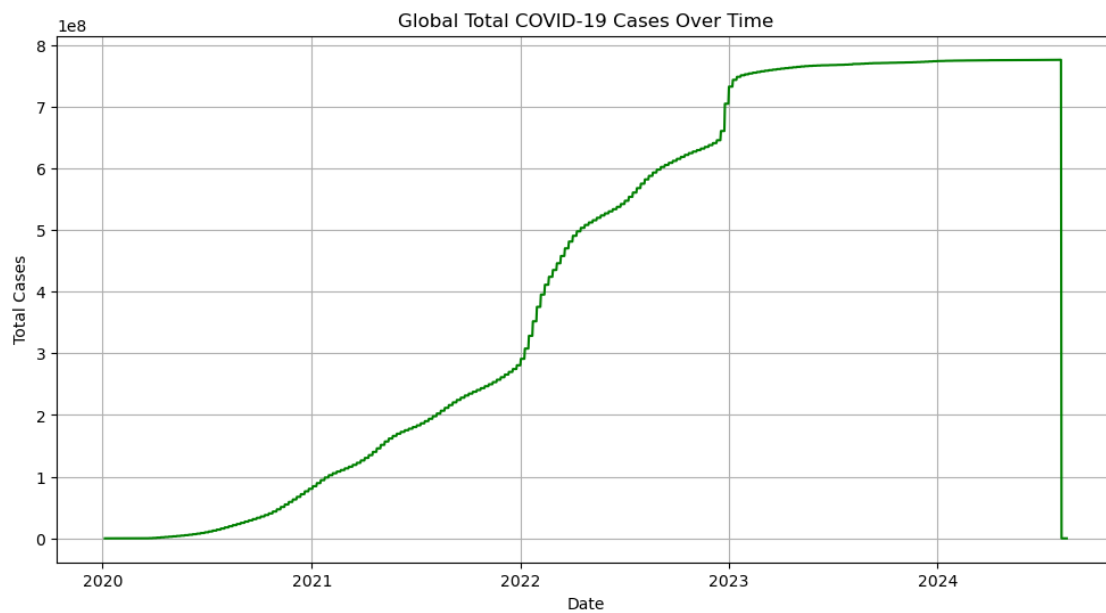
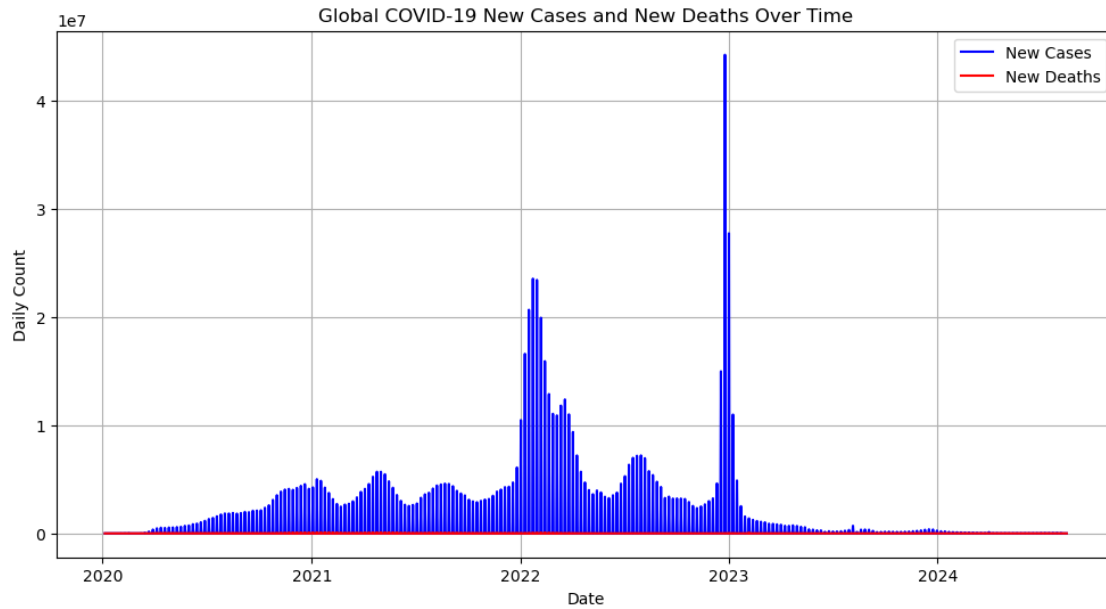
Date converted + missing values filled with 0 in key columns!

Missing values left in main columns:

```
date            0
location        0
new_cases       0
total_cases     0
new_deaths      0
total_deaths    0
dtype: int64
```

```
[5]: global_df = df[df['location'] == 'World'].copy()
plt.figure(figsize=(12,6))
sns.lineplot(x='date', y='new_cases', data=global_df, label='New Cases',
             color='blue')
sns.lineplot(x='date', y='new_deaths', data=global_df, label='New Deaths',
             color='red')
plt.title('Global COVID-19 New Cases and New Deaths Over Time')
plt.xlabel('Date')
plt.ylabel('Daily Count')
plt.legend()
plt.grid(True)
plt.show()
plt.figure(figsize=(12,6))
sns.lineplot(x='date', y='total_cases', data=global_df, color='green')
plt.title('Global Total COVID-19 Cases Over Time')
plt.xlabel('Date')
plt.ylabel('Total Cases')
plt.grid(True)
plt.show()

print(" Two time-series plots created successfully!")
print("You can clearly see the waves of COVID spread!")
```



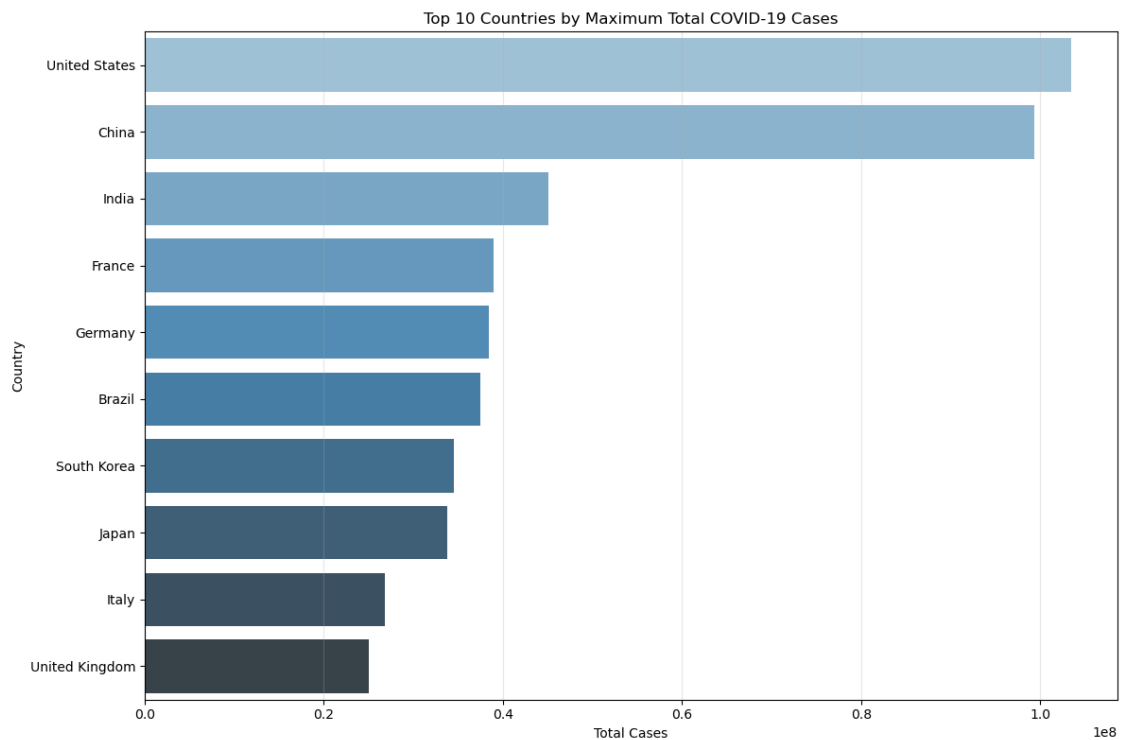
Two time-series plots created successfully!
You can clearly see the waves of COVID spread!

```
[9]: country_max = df.groupby('location').agg({
    'total_cases': 'max',
    'total_deaths': 'max',
    'continent': 'first'
```

```

}).reset_index()
real_countries = country_max[country_max['continent'].notna()]
top10 = real_countries.nlargest(10, 'total_cases')
plt.figure(figsize=(12, 8))
sns.barplot(y='location', x='total_cases', data=top10, palette='Blues_d')
plt.title('Top 10 Countries by Maximum Total COVID-19 Cases')
plt.xlabel('Total Cases')
plt.ylabel('Country')
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()
print(" Top 10 Countries by Highest Total Cases Ever:")
print(top10[['location', 'total_cases', 'total_deaths']].round(0).
      ↪to_string(index=False))
print("\n Reliable country-wise comparison plot created successfully!")

```



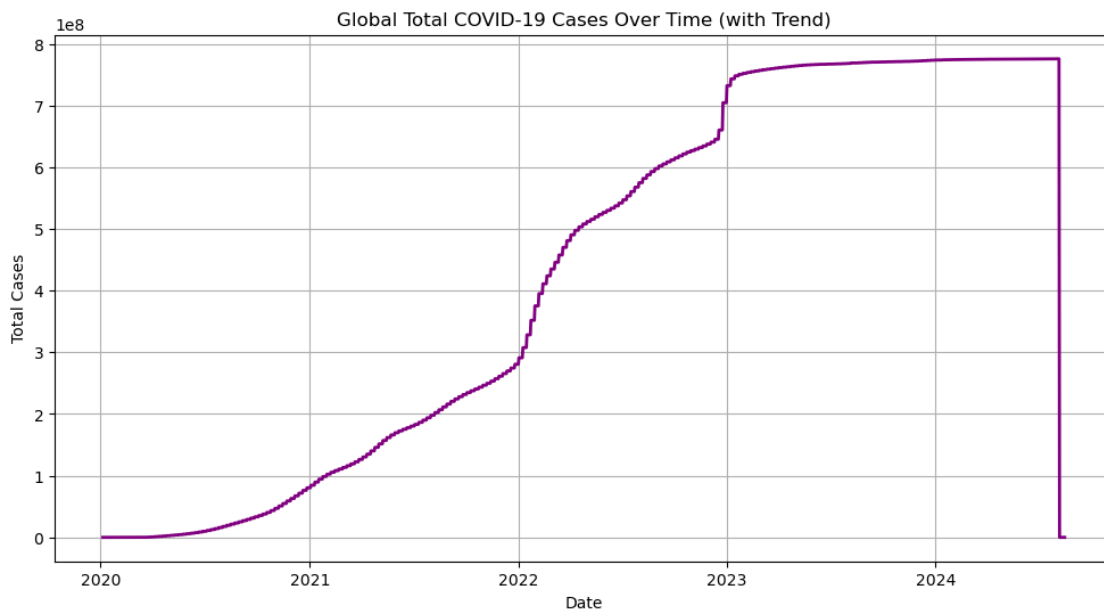
Top 10 Countries by Highest Total Cases Ever:

location	total_cases	total_deaths
United States	103436829.0	1193165.0
China	99373219.0	122304.0
India	45041748.0	533623.0
France	38997490.0	168091.0
Germany	38437756.0	174979.0
Brazil	37511921.0	702116.0

South Korea	34571873.0	35934.0
Japan	33803572.0	74694.0
Italy	26781078.0	197307.0
United Kingdom	24974629.0	232112.0

Reliable country-wise comparison plot created successfully!

```
[10]: import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(12,6))
sns.lineplot(x='date', y='total_cases', data=global_df, color='purple',
             linewidth=2)
plt.title('Global Total COVID-19 Cases Over Time (with Trend)')
plt.xlabel('Date')
plt.ylabel('Total Cases')
plt.grid(True)
plt.show()
recent = global_df.tail(30)
growth_rate = (recent['new_cases'].mean() / global_df['total_cases'].max()) * 100
print(" Simple Forecasting Insight:")
print(f"Average daily new cases in last 30 days: {recent['new_cases'].mean():.0f}")
print(f"Growth rate compared to total cases: {growth_rate:.3f}% per day")
```



Simple Forecasting Insight:
Average daily new cases in last 30 days: 5175

Growth rate compared to total cases: 0.001% per day

```
[ ]: S
```